

Which Generators Retire?

Application of Machine Learning Models to Generator Data from the
PJM Footprint

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Term Project

ECON 685: Economic Applications of Machine Learning with Python

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Summer 2026

Data source: U.S. Energy Information Administration, Form EIA-860 (2024 release). This project uses raw public data rather than a published article's dataset.

Abstract

I analyze which power generating units retire, using the 2024 EIA-860 generator survey: 7,660 units in the 13 PJM-footprint states, 16.4 percent retired. I apply a logistic model with threshold selection, forward stepwise and L1 model selection, a Generalized Additive Model using B-splines of age and capacity, and Random Forest, Gradient Boosting, and XGBoost models. Using the predicted probabilities of the logistic model, the thresholds that maximize accuracy, the F-score, and the kappa index in the test set all occur at 0.42. The performance measures indicate that the Random Forest outperforms the others, and retired units are old, small, and fuel-inflexible.

I. Introduction

The PJM Interconnection operates the largest wholesale electricity market in the United States, and its footprint has become the center of data-center-driven electricity demand growth. At the same moment demand is accelerating, retirements keep subtracting from the existing fleet: units leave every year, and the ones that leave are not a random sample. Understanding what retired units have in common is a first step toward understanding where the operating fleet is thin. This paper asks a deliberately narrow question: given a generating unit's physical and operational attributes, how closely does it match the profile of units that have already retired?

The economics literature on plant retirement emphasizes market forces and regulation. Fell and Kaffine (2018) show that low natural gas prices and growing renewable generation jointly drove coal generation down, and Davis, Holladay, and Sims (2022) document that retiring coal units are disproportionately old, small, and inefficient. My contribution here is not to advance that literature but to test how much of the retirement pattern is recoverable from physical attributes alone, using a standard machine learning toolkit: classification models, threshold and model selection, nonlinear specifications, and tree ensembles, evaluated out of sample following James et al. (2023) and Lantz (2019).

The main findings are easy to state. Retirement status is highly learnable: the logistic model reaches a test AUC of 0.92, and the random forest reaches 0.98 with a test accuracy of 0.96 and an F-score of 0.87 at the default threshold. On the logit model, the thresholds that maximize accuracy, the F-score, and Cohen's kappa in the test set all land at 0.42. The retired profile is old, small, and inflexible, which matches the coal-retirement literature. Two results sharpen the picture. First, forward selection under a BIC criterion keeps only 8 of 18 variables at essentially full-model performance, so the signal is concentrated. Second, although age is the single strongest predictor, removing it from the full models costs only about 0.015 AUC: fuel, technology, and unit size carry most of the retirement signal independently of vintage.

II. Data

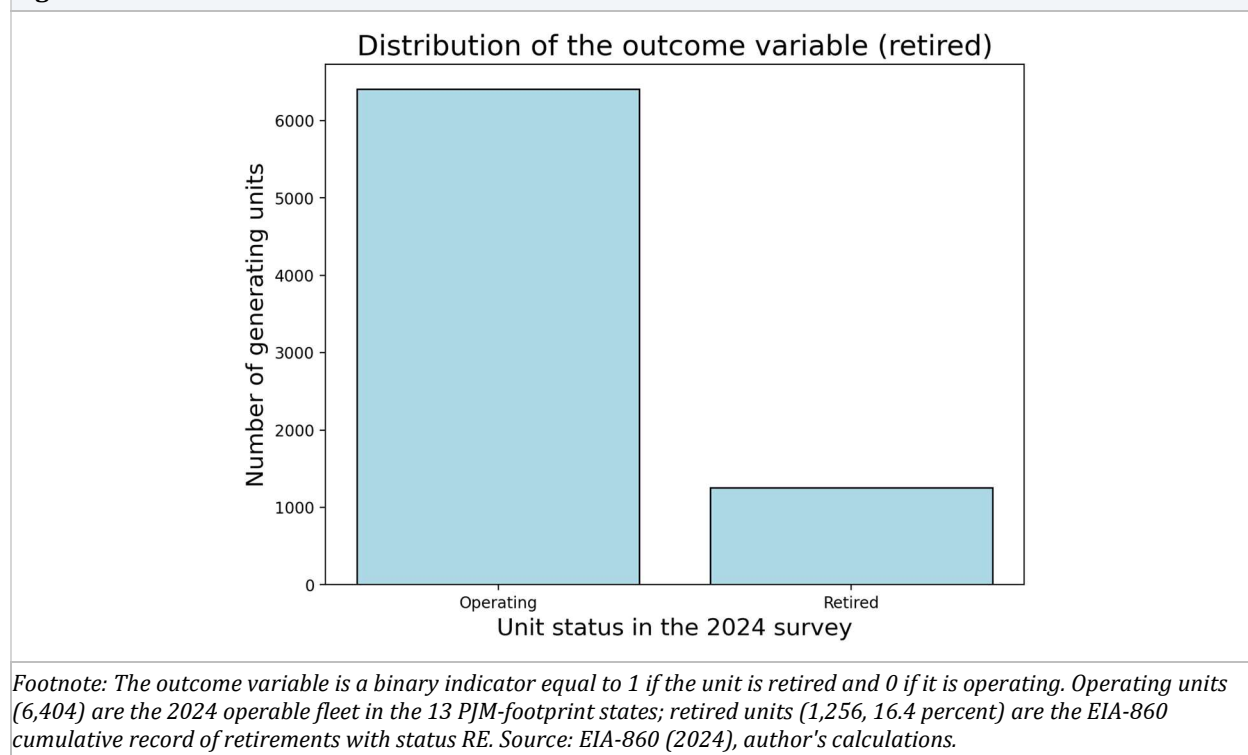
The data come from Form EIA-860, the U.S. Energy Information Administration's annual census of generating units, 2024 release (EIA, 2024). The unit of observation is the generating unit, identified by plant code and generator ID. I use the generator-level workbook, which reports each unit's capacity, technology, fuels, vintage, and status. Because the survey is a census rather than a sample,

coverage of the fleet is essentially complete. The workbook arrives as raw survey output, and the data preparation described below starts from that unprocessed file.

I construct the outcome by stacking two sheets of the workbook. Units on the Operable sheet form the operating class (retired = 0). Units on the Retired and Canceled sheet with status code RE, meaning genuinely retired rather than canceled or postponed, form the retired class (retired = 1). I restrict both classes to the 13 states of the PJM footprint. The result is 7,660 units, of which 1,256 (16.4 percent) are retired. The retired class is the survey's cumulative record of retirements through 2024, while the operating class is a 2024 snapshot; the dataset is a cross section, not a time series, so the analysis characterizes fleet composition rather than forecasting closure dates.

Retirement status is the right outcome for the question because it is the administrative fact of a unit leaving the fleet, recorded by the same survey that records the predictors, and its binary form makes the full classification toolkit applicable. Figure 1 shows its distribution: the outcome is imbalanced, which shapes the threshold analysis in Section III.A.

Figure 1: Distribution of the Outcome Variable



Data preparation involves four steps. First, I engineer two variables: age, defined as 2024 minus the unit's initial operating year, and multi_fuel, a flag equal to one when the unit reports any secondary energy source. Both are available on both sheets, so neither leaks the outcome. Second, two numeric fields, minimum load and nameplate power factor, are missing for roughly a third of units; I add missingness indicator variables and then impute medians, so the fact of missingness remains visible to the models. Third, I collapse categorical levels with fewer than 30 units into an Other category, which stabilizes the logistic regression against quasi-separation. Fourth, I drop three technology flags

that are nearly constant across the sample. The final table has 20 candidate predictors: six continuous, three binary numeric, and eleven categorical.

The predictors fall into five economic categories: vintage (age), physical size (nameplate, summer, and winter capacity, minimum load), technology and fuel (technology class, prime mover, primary energy source, combustion-technology flags, cofiring, fuel flexibility), operational characteristics (power factor and the missingness indicators), and location and ownership (state, sector). Each category has a plausible link to retirement economics: older vintages embody obsolete technology, small units spread fixed costs over less output, fuel-inflexible units cannot arbitrage fuel prices, and sector determines exposure to market discipline. A complete list appears in Appendix A (Tables A1 and A2).

III. Analysis

A. Logit Model and Thresholds

I split the data into a 75 percent training set and a 25 percent test set before any design matrix is fit, so no information from the test set enters the transformations. The baseline model is a logistic regression (a GLM with a binomial family) on 18 variables; summer and winter capacity are excluded from the linear specifications because they correlate with nameplate capacity at roughly 0.99, and they are retained for the tree models, which collinearity does not affect. The logit's overall validity is checked by comparing training and test performance: the training AUC is 0.929 and the test AUC is 0.924, and accuracy at the default 0.50 threshold is 0.907 in training and 0.906 in the test set. The near-zero gap means the model is not overfitting, which is expected given 5,745 training observations against 68 design columns.

The most significant predictors tell a coherent story. Age enters positively (coefficient 0.046, $z = 11.3$): each additional year of vintage raises the odds of retirement by roughly 4.7 percent. Nameplate capacity enters negatively: larger units retire less, consistent with their lower average costs and stronger capacity-market position. The multi_fuel flag is negative, so fuel flexibility is protective. The missingness indicator for minimum load is strongly positive. This is best read as a record-keeping artifact rather than an economic effect: older units have less complete survey records, so a blank field is itself a marker of vintage. Among the variables that lack significance, most state dummies and several sector levels contribute little once technology and vintage are controlled for, which suggests retirement is driven by what a unit is rather than where it is.

With 16.4 percent of units retired, a classifier that predicts operating for every unit is already correct 83 percent of the time, so the default 0.50 cutoff carries no special authority. I sweep thresholds from 0.01 to 0.89 and compute accuracy, recall, precision, the F-score, and Cohen's kappa in the test set at each step (Appendix Figure B1). The thresholds that maximize accuracy, the F-score, and kappa all occur at 0.42, an unusually clean agreement across the three criteria. At 0.42 the test-set confusion table gives accuracy 0.911, precision 0.834, recall 0.583, and an F-score of 0.686; at the default 0.50 the same model gives accuracy 0.906, precision 0.879, recall 0.502, and an F-score of 0.639. The comparison shows what the threshold buys: moving from 0.50 to 0.42 trades a little precision for a meaningful gain in recall. Whether that trade is worth making depends on the loss function: a planner

worried about missing fragile capacity values recall, while a researcher flagging units for costly case studies values precision. Since neither cost is stated here, I treat the F-score's balance as the headline and use the default 0.50 threshold for all subsequent models so their measures are directly comparable.

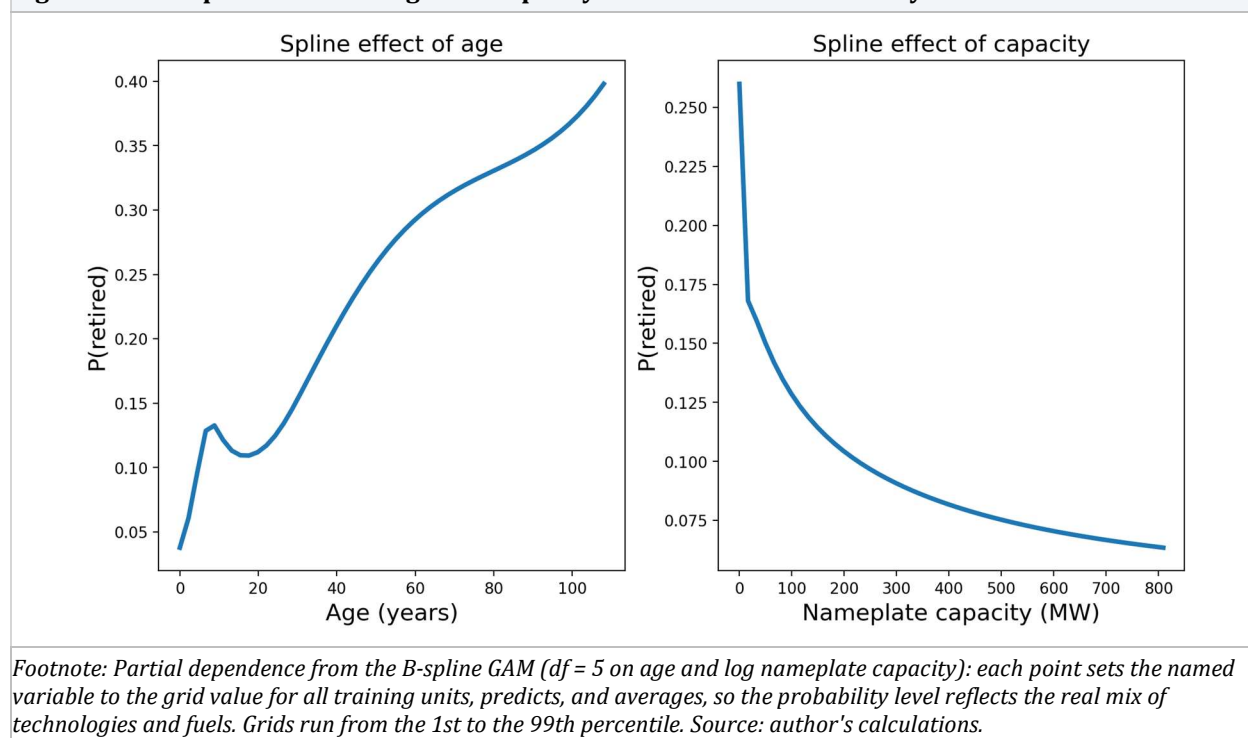
B. Model Selection

I apply two model selection methods to the logistic specification. First, forward stepwise selection, with the BIC criterion adapted to classification by replacing the residual sum of squares with the binomial deviance. Forward selection keeps 8 of 18 variables: age, nameplate capacity, technology, power factor, multi_fuel, the fluidized bed flag, and the two missingness indicators. The design shrinks from 68 columns to 23, and the test AUC moves from 0.924 to 0.922, a loss of two thousandths. Second, an L1-penalized logistic regression, the classification analogue of the lasso, estimated in a pipeline that standardizes the design and chooses the penalty by 5-fold cross-validation on the training set. The L1 model keeps 58 of 67 columns and posts a test AUC of 0.925. The BIC's $\log(n)$ penalty prunes far more aggressively than the cross-validated lasso, which is the expected ordering: both methods agree the model can shed complexity at no material cost. Section III.E returns to this result.

C. Non-linear Model

The GAM replaces the linear terms for the two central continuous predictors with B-splines of five degrees of freedom each: age, and the log of nameplate capacity, where the log scale spreads the spline knots evenly across a variable spanning four orders of magnitude. All other variables stay linear. A nonlinear specification is appropriate for these two variables on economic grounds: retirement risk should not rise at a constant rate over a century of vintages, and the difference between a 5 MW and a 100 MW unit should matter more than the difference between 700 and 800 MW. The GAM's test AUC is 0.929 against the linear logit's 0.924, and its accuracy, recall, and F-score at the 0.50 threshold are 0.911, 0.542, and 0.671, each above the linear logit's, with precision essentially unchanged (0.878 against 0.879). The spline terms therefore improve the model, modestly but consistently.

Figure 2: GAM Spline Effects of Age and Capacity on Retirement Probability



Both curves are economically sensible. Retirement probability climbs steeply in age, from about 4 percent for new units to about 40 percent at the oldest vintages, with a plateau through middle age and a small bump between ages 5 and 10 that plausibly reflects early-life attrition of units that never performed as intended. Capacity works in the opposite direction: risk concentrates below roughly 100 MW and flattens beyond, so the protective effect of size is mostly about not being small rather than being enormous. These are exactly the shapes a linear logit cannot represent, which is where the GAM's gain comes from.

D. Tree Models

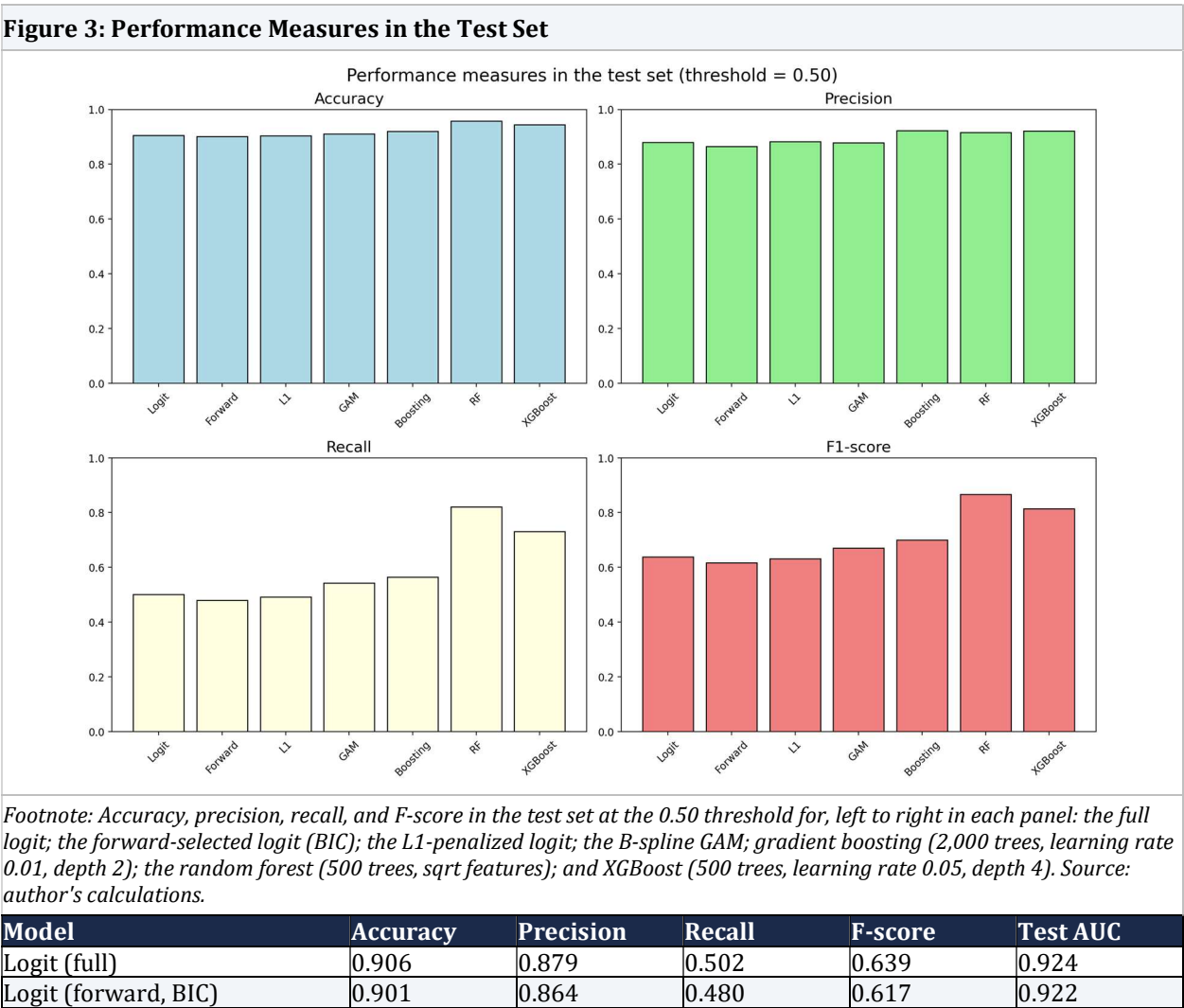
I estimate three tree ensembles on the full 20-variable design. The random forest grows 500 trees, each split considering a random square-root subset of the features, and predicts by averaging; decorrelating the trees through feature sampling is what separates it from simple bagging. The gradient boosting model builds 2,000 depth-2 trees in sequence with a learning rate of 0.01, each tree fitted to the errors the ensemble is still making; the small learning rate and shallow trees are the standard guard against overfitting. XGBoost, the advanced third model, uses 500 depth-4 trees with a learning rate of 0.05 and row and column subsampling (90 and 80 percent), which adds further regularization. Class imbalance is handled through the threshold analysis of Section III.A rather than reweighting.

On the test set at the 0.50 threshold, the random forest posts accuracy 0.958, precision 0.916, recall 0.821, and an F-score of 0.866; XGBoost posts 0.945, 0.921, 0.730, and 0.815; and gradient boosting posts 0.920, 0.923, 0.564, and 0.700. The boosting model's combination of the highest precision (0.923) with much lower recall reflects its conservative configuration: depth-2 trees with a small learning rate produce probabilities that cross 0.50 only for the clearest retirement profiles. The

random forest's training AUC of 0.9996 against its test AUC of 0.979 reflects near-memorization of the training data, which is expected behavior for deep forests; performance claims for such models are only meaningful on held-out data. Variable importance graphs for all three models appear in Appendix Figures B2 through B4, and their comparison is taken up in the next section.

E. Model Comparison and Performance Evaluation

Figure 3 collects accuracy, precision, recall, and the F-score for all seven models in the test set at the common 0.50 threshold, and Table 1 reports the same values along with each model's AUC. Three patterns organize the comparison. First, accuracy and precision are nearly flat across models: every model sits between 0.90 and 0.96 accuracy and between 0.86 and 0.92 precision, so neither measure separates the field. Second, recall is where the models differ: the linear family and the GAM find 48 to 54 percent of retired units at 0.50, while XGBoost finds 73 percent and the random forest 82 percent. Third, the F-score therefore ranks the ensembles first (random forest 0.866, XGBoost 0.815), the GAM in the middle (0.671), and the linear logits together at the bottom (0.62 to 0.64). The AUC column tells the same story free of any threshold: 0.92 to 0.93 for the linear family and GAM, 0.955 for boosting, 0.974 for XGBoost, and 0.979 for the random forest.



Logit (L1)	0.904	0.882	0.492	0.632	0.925
GAM (B-spline)	0.911	0.878	0.542	0.671	0.929
Gradient boosting	0.920	0.923	0.564	0.700	0.955
Random forest	0.958	0.916	0.821	0.866	0.979
XGBoost	0.945	0.921	0.730	0.815	0.974

Table 1: Model comparison in the test set (threshold = 0.50). Footnote: all models fit on the same 75/25 train/test split (random_state = 0). ROC curves appear in Appendix Figure B6. Source: author's calculations.

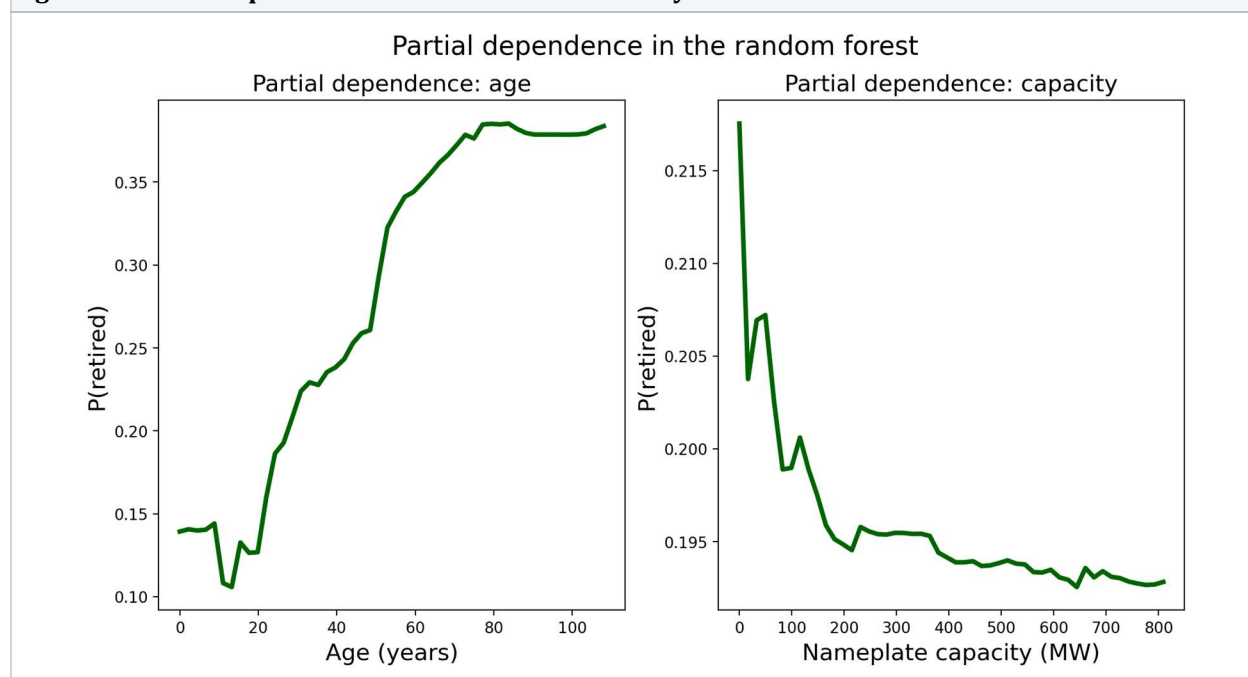
The explanation for the ranking is interactions. The GAM already captures the univariate nonlinearity in age and capacity, yet the ensembles still open a gap of four to five points of AUC, so the remaining signal must live in combinations of predictors: old and small and coal-adjacent describes a specific joint profile that trees capture natively while additive models cannot. Each family also has its advantages: the logit is transparent and its coefficients have odds interpretations; the GAM adds interpretable nonlinear shapes at modest cost; the ensembles buy the best prediction at the cost of interpretability, which the importance graphs and partial dependence plots only partially recover.

The importance rankings across tree models agree on the core and disagree instructively at the margins. The random forest's impurity-based ranking (Figure B2) is led by age, the capacity fields, the minimum-load missingness indicator, and power factor. Gradient boosting (Figure B3) concentrates its importance on age and the conventional steam coal technology dummy. XGBoost's gain-based ranking (Figure B4) elevates the missingness flag and prime mover dummies while pushing age out of its top ten entirely. The disagreement is methodological rather than substantive: impurity importance favors continuous variables that offer many split points, while gain importance rewards dummies that split cleanly. Read together, the models consistently identify vintage, size, and coal-adjacent technology as the retirement profile.

On the effectiveness of model selection: forward selection under BIC cut the design from 68 columns to 23, a two-thirds reduction in model size, while moving test AUC only from 0.924 to 0.922, and the L1 penalty kept 58 of 67 columns at 0.925. Model selection therefore succeeds on both fronts that matter: it demonstrates that the predictive signal is concentrated in a small set of variables, and it delivers a far simpler model with no meaningful performance cost. If the goal were a model to explain to a regulator, the 23-column forward model is the one to carry.

Figure 4 closes the evaluation with partial dependence plots for the two most influential continuous variables, constructed from the best-performing model, the random forest. The age curve rises from roughly 14 percent to 38 percent across the observed range, with the same early-life dip-and-climb the GAM found. The capacity curve falls as expected but moves in a much narrower band than the GAM's: forest averaging across 500 trees flattens marginal effects, and part of capacity's influence is absorbed by its near-duplicates (summer and winter capacity) that remain in the tree design. The two figures agree on direction and disagree on amplitude, and the amplitude difference is roughly the portion of the size effect that nameplate capacity shares with its near-duplicate measures: the GAM assigns the whole effect to one variable, while the forest divides it among three.

Figure 4: Partial Dependence of Retirement Probability in the Random Forest



Footnote: Partial dependence computed by setting the named variable to each grid value for every training unit, predicting with the fitted 500-tree random forest, and averaging. Grids run from the 1st to the 99th percentile. The narrower vertical range relative to Figure 2 reflects averaging across trees and the presence of correlated capacity measures in the tree design. Source: author's calculations.

Taken together, the findings are these: retirement status in the PJM fleet is highly predictable from physical and operational attributes alone, with test AUC between 0.92 and 0.98 across the seven models; the retired profile is old, small, and fuel-inflexible; the predictive signal is concentrated in roughly eight variables; and the ensembles' advantage comes from interactions among predictors rather than from nonlinearity in any single one.

F. Extensions

Because age is the most obvious retirement predictor, the natural objection is that every model above is an age model in disguise. An ablation answers it. A logistic regression with age as the only predictor reaches a test AUC of 0.79: far above chance, far below the full models. Removing age from the full models costs 0.014 AUC for the logit, 0.011 for the random forest, and 0.021 for XGBoost, an average of about 0.015 (Table 2). Age is sufficient for a decent model but necessary for none: fuel, technology, and size carry most of the signal independently of vintage.

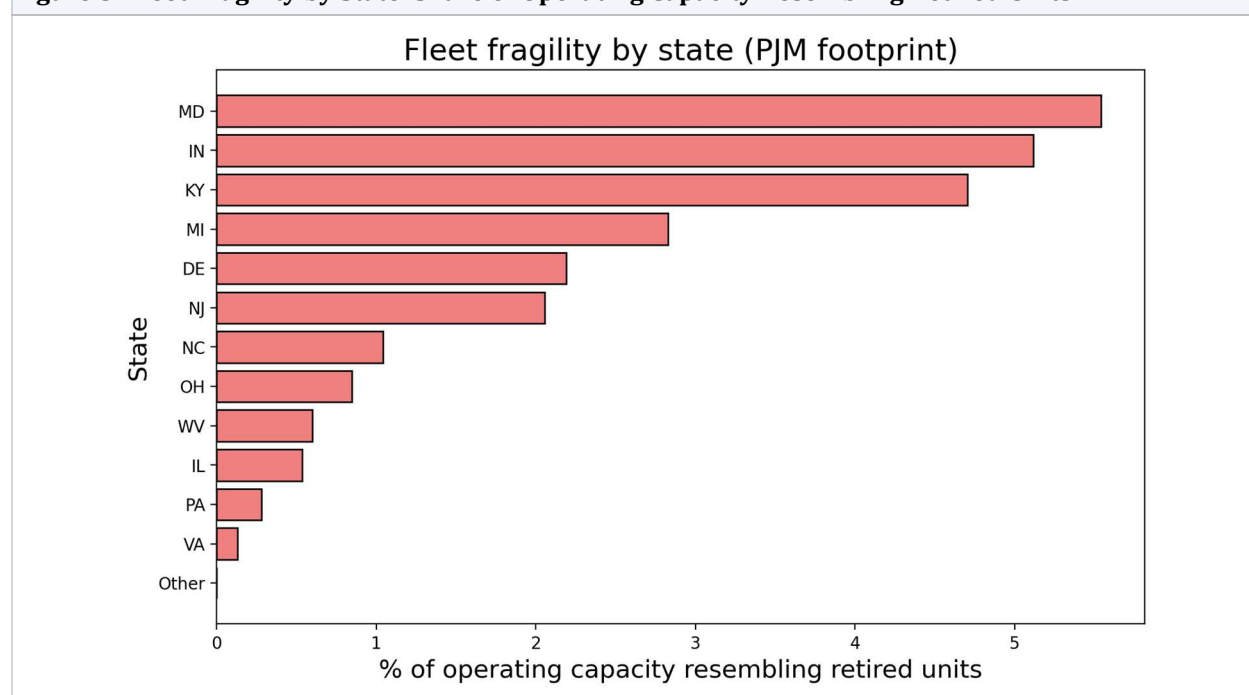
Model	AUC with age	AUC without age	Drop
Logistic	0.924	0.910	0.014
Random forest	0.979	0.969	0.011
XGBoost	0.974	0.954	0.021

Table 2: Age ablation, test AUC. Footnote: the age-only logistic regression reaches a test AUC of 0.793. Source: author's calculations.

The second extension turns the classifier on the operating fleet. I score every unit with an out-of-fold retirement-resemblance probability from a random forest under 5-fold cross-validation, so no unit

is scored by a model trained on it. Operating units whose score exceeds the Section III.A threshold of 0.42 count as resembling retired units, and I aggregate the capacity-weighted share by state. Figure 5 shows the result: Maryland, Indiana, and Kentucky hold the largest shares of retirement-resembling operating capacity, around 5 percent each, while Pennsylvania and Virginia hold less than half a percent. This is a fleet-composition lens rather than a closure forecast, but it points at where the operating fleet most resembles the units the market has already pushed out.

Figure 5: Fleet Fragility by State: Share of Operating Capacity Resembling Retired Units



Footnote: Out-of-fold random forest resemblance scores (5-fold stratified CV); operating units with score at or above 0.42 count as fragile; shares are capacity-weighted by nameplate MW. DC and collapsed small-state levels appear as Other. Source: author's calculations.

IV. Concluding Remarks

This paper asked whether a generating unit's physical and operational attributes identify it as retired or operating, using 7,660 units in the PJM footprint from the 2024 EIA-860 survey. The answer is emphatically yes: all seven models clear a test AUC of 0.92, and the random forest reaches 0.98 with a test F-score of 0.87. The retired profile that emerges, old, small, and fuel-inflexible, is consistent with the retirement literature, and the ablation shows the profile is not reducible to age. Model selection concentrates the signal into eight variables at essentially no performance cost, and the fragility extension shows how the fitted model can characterize today's operating fleet state by state.

The limitations are inherent to the design. The data are a cross-sectional snapshot: the retired class accumulates over decades while the operating class is observed in 2024, so the model characterizes what retired units look like rather than when any unit will retire, and survivor composition differs across vintages. The models also see no economics: no fuel prices, no capacity-market revenues, no environmental compliance costs, all of which the literature identifies as retirement drivers. Future

work should move to the panel version of EIA-860, where retirement dates support survival analysis, and merge in market variables so the physical profile and the economic drivers can compete in one model.

The practical contribution is the resemblance score itself. For planners in a region where large new loads are arriving quickly, a capacity-weighted map of which operating units most resemble past retirements is a cheap, transparent screening tool: it does not predict closures, but it identifies where the fleet is oldest, smallest, and least flexible, which is where supply-side surprises are most likely to originate.

V. References

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VI. Appendix A: Variable Description

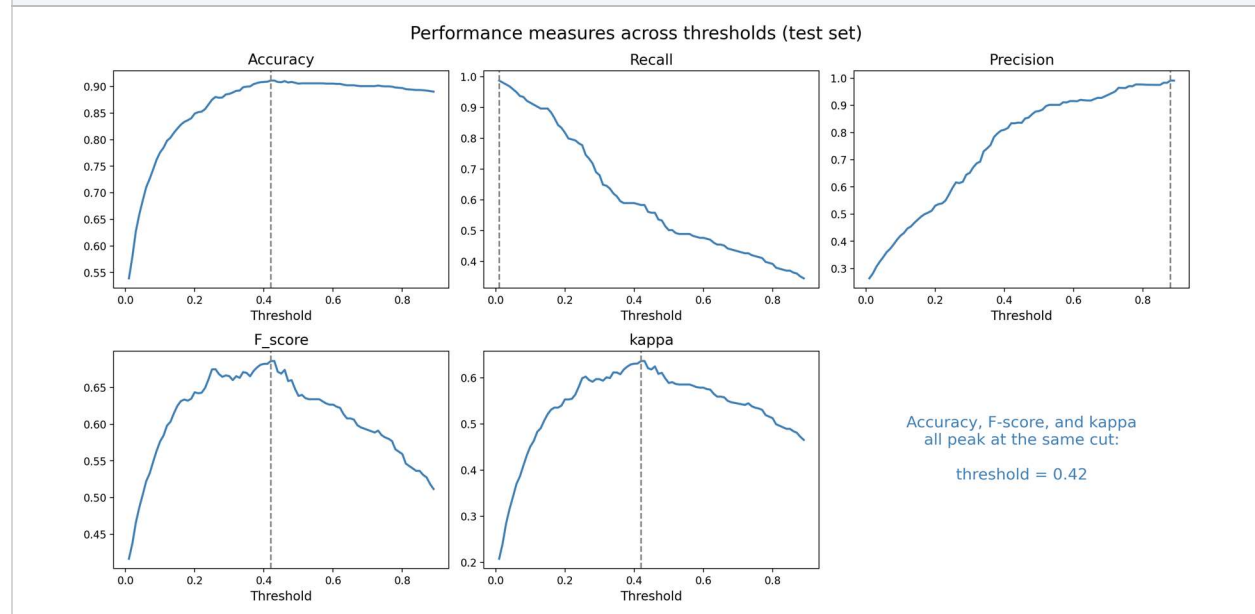
Table A1: Dependent Variable	
retired	Binary indicator equal to 1 if the unit appears on the EIA-860 Retired and Canceled sheet with status RE (genuinely retired), and 0 if the unit appears on the Operable sheet. Restricted to the 13 PJM-footprint states.

Table A2: Explanatory Variables	Description
age	Numerical (engineered): 2024 minus the unit's initial operating year.
Nameplate Capacity (MW)	Numerical: maximum rated output of the generator.
Summer Capacity (MW)	Numerical: net summer capability. Tree models only (collinear with nameplate).
Winter Capacity (MW)	Numerical: net winter capability. Tree models only (collinear with nameplate).
Minimum Load (MW)	Numerical: minimum stable output. Median-imputed; see missingness flag.
Nameplate Power Factor	Numerical: rated power factor. Median-imputed; see missingness flag.
multi_fuel	Binary (engineered): 1 if any secondary energy source (Energy Source 2 through 6) is reported.
Minimum Load (MW)_missing	Binary (engineered): 1 if minimum load was missing before imputation (about one third of units).
Nameplate Power Factor_missing	Binary (engineered): 1 if power factor was missing before imputation (about one third of units).
State	Categorical: state of the plant (13 PJM-footprint states); rare levels collapsed to Other.
Technology	Categorical: EIA technology class (e.g., Conventional Steam Coal, Solar Photovoltaic); rare levels collapsed to Other.
Prime Mover	Categorical: prime mover code (e.g., ST steam turbine, CT combustion turbine); rare levels collapsed.
Energy Source 1	Categorical: primary fuel code; rare levels collapsed to Other.
Sector Name	Categorical: ownership sector (electric utility, IPP, commercial, industrial).
Cofire Fuels?	Categorical flag: Y if the unit can cofire multiple fuels; N otherwise.
Fluidized Bed Technology?	Categorical flag: Y if fluidized bed combustion; N otherwise.
Pulverized Coal Technology?	Categorical flag: Y if pulverized coal combustion; N otherwise.
Stoker Technology?	Categorical flag: Y if stoker combustion; N otherwise.
Subcritical Technology?	Categorical flag: Y if subcritical steam cycle; N otherwise.
Supercritical Technology?	Categorical flag: Y if supercritical steam cycle; N otherwise.

Footnote: three additional flags (Carbon Capture, Ultrasupercritical, Solid Fuel Gasification) were dropped as near-constant (minority level below 30 units). Source: EIA-860 (2024).

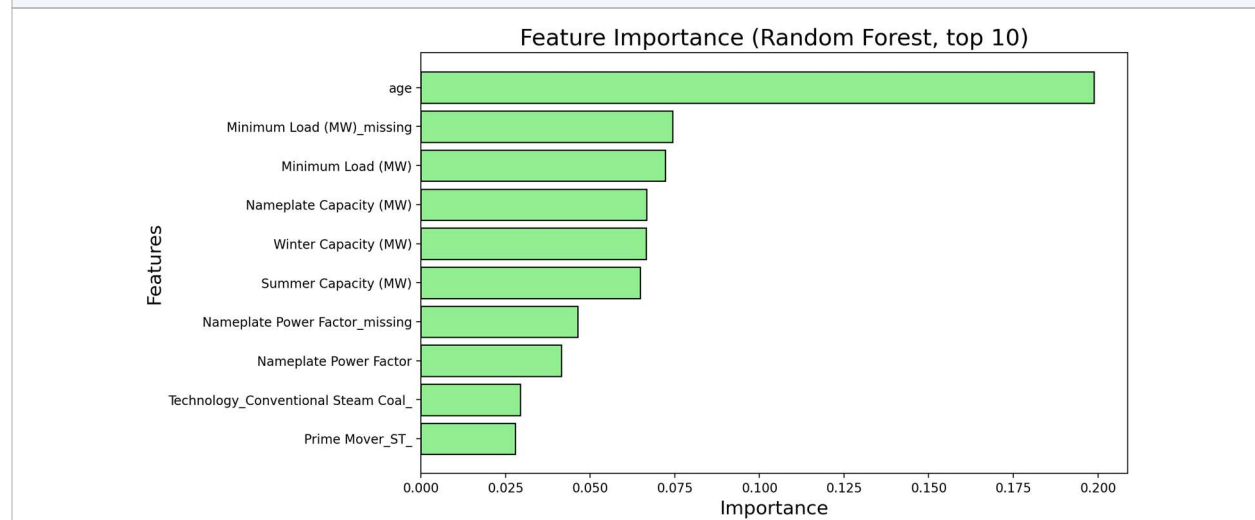
Appendix B: Graphs

Figure B1: Performance Measures as a Function of the Threshold



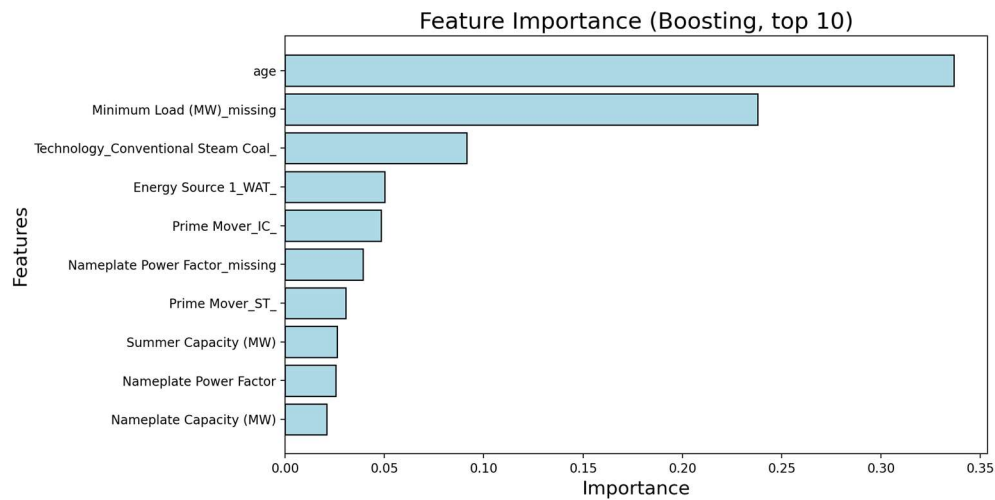
Footnote: Accuracy, recall, precision, F-score, and Cohen's kappa in the test set for the full logit model, across thresholds from 0.01 to 0.89 in steps of 0.01. The vertical line in each panel displays the threshold with the maximum value of that performance measure; accuracy, F-score, and kappa all peak at 0.42. Source: author's calculations.

Figure B2: Variable Importance: Random Forest



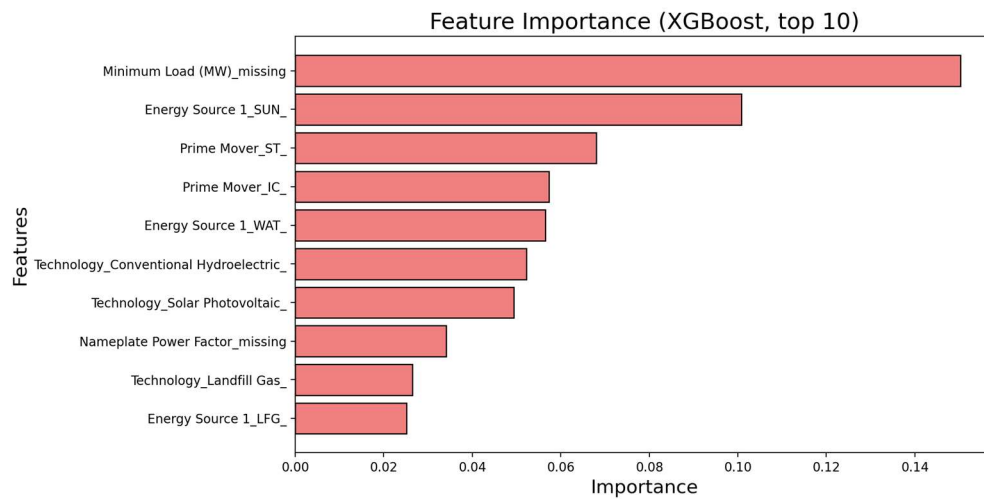
Footnote: Relative feature importance (mean decrease in impurity) of the top 10 design columns across 500 trees. Categorical variables appear as individual dummy columns. Source: author's calculations.

Figure B3: Variable Importance: Boosting Tree



Footnote: Relative feature importance of the top 10 design columns in the gradient boosting model (2,000 trees, learning rate 0.01, depth 2). Source: author's calculations.

Figure B4: Variable Importance: XGBoost Tree

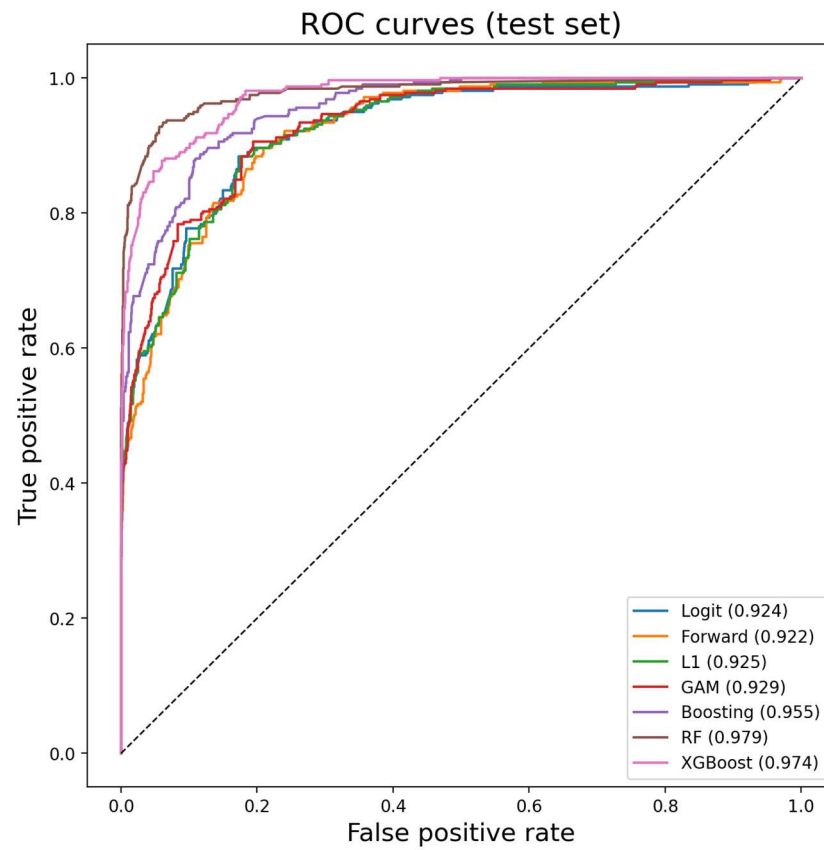


Footnote: Gain-based relative feature importance of the top 10 design columns. Gain rewards dummies that split cleanly (the missingness flag, prime movers), while the impurity measure of Figure B2 favors continuous variables with many split points; the two views are complementary. Source: author's calculations.

Figure B5: Age Distribution by Retirement Status



Figure B6: ROC Curves in the Test Set



Footnote: Receiver operating characteristic curves for all seven models on the held-out test set (1,915 units); legend entries show test AUC. The dashed diagonal is the no-information benchmark. Source: author's calculations.